

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$S_{\text{off}}^{c \geq p^2} := \sum_{d_1, d_2 < D} \sum_{Id_1, Jd_2 < D} x_{Id_1} \bar{x}_{Jd_2} \sqrt{IJ} \sum_{L, M \geq 1} \frac{G_k(LM d_1 d_2 / p)}{\sqrt{LM}} \\ \times \left(2\pi i^{2k} \sum_{\substack{p|c \\ c \geq p^2}} \frac{\mathcal{S}(IL, JM, c)}{c} J_{2k-1} \left(\frac{4\pi \sqrt{ILJM}}{c} \right) \right).$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} 0 < \kappa_\nu(u) &= \det(f_u(u, 0), f_{vv}(u, 0), f_{uu}(u, 0)) \\ &= \det(T \circ \tilde{f}_u(u, 0), T \circ \tilde{f}_{vv}(u, 0), T \circ \tilde{f}_{uu}(u, 0)) \\ &= \det(\tilde{f}_u(u, 0), \tilde{f}_{vv}(u, 0), \tilde{f}_{uu}(u, 0)) \\ &= \xi_u(u, 0) \det(\check{f}_\xi(\xi(u, 0), 0), \check{f}_{\eta\eta}(\xi(u, 0), 0), \check{f}_{\xi\xi}(\xi(u, 0), 0)). \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{Q}_t(x_{1:t-1}) = \rho_t \left(\mathfrak{Q}_t(x_{1:t-1}, \xi_t) \right) &\geq \rho_t \left(\mathfrak{Q}_t^{k-1}(x_{1:t-1}, \xi_t) \right) \\ &\geq \sup_{p \in \mathcal{P}_t} \sum_{j=1}^M p_j \Phi_{t,j} \mathfrak{Q}_t^{k-1}(x_{1:t-1}, \xi_{t,j}). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} z_1(\xi) &= r_\xi \left(\omega_\xi^* - \frac{\alpha \xi^\alpha}{2(1-\alpha)} \right) \\ &= r_\xi \left(\frac{1 - \alpha \xi - \frac{\alpha}{2} \xi^\alpha}{1-\alpha} \right) \\ &= \dots = \frac{1}{(1-\alpha)^{1-\alpha} \alpha^\alpha} \frac{\xi - 1 + \frac{\alpha}{2} \xi^\alpha}{(\xi - 1 + \frac{1}{2} \xi^\alpha)^\alpha}, \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
& \frac{\partial}{\partial t} u_{ij}(t; x_1, \dots, x_k; y_1, \dots, y_k) \\
&= \sum_{i'=1}^k \sum_{j'=1}^k \frac{\pi_{i'j'} x_{i'} y_{j'}}{n} [u_{ij}(t; x_1 - \delta_{1i'}, \dots, x_k - \delta_{ki'}, y_1 - \delta_{1j'}, \dots, y_k - \delta_{kj'}) + \delta_{ii'} \delta_{jj'}] \\
&\quad - \frac{z}{n} u_{ij}(t; x_1, \dots, x_k; y_1, \dots, y_k).
\end{aligned}$$